EM Algorithm and Newton-Raphson Method for Imputing Missing Data in Healthcare Records: Joint Modeling of Longitudinal and Time-to-Event Data

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The exponential growth of healthcare data, driven by advancements in electronic health records (EHR), wearable technologies, mobile health applications, and clinical trials, has transformed the landscape of patient care and medical research. These data sources have enabled researchers and clinicians to analyze large volumes of information to identify trends, predict outcomes, and personalize treatments. However, healthcare datasets are inherently complex, heterogeneous, and often incomplete, posing significant challenges for effective analysis and interpretation.Missing data is one of the most pervasive issues in healthcare analytics. It can occur for various reasons, such as:Patients may discontinue participation in clinical trials or regular follow-ups, resulting in incomplete longitudinal measurements. Manual errors, system inefficiencies, or unstructured data formats in EHRs can lead to missing entries in critical variables such as lab results or treatment details.Technical issues with wearable devices or diagnostic equipment can cause gaps in the recorded data. These missing values, if left unaddressed, can bias statistical analyses, leading to misleading results and suboptimal decisions in patient care. For example, biased risk estimates in survival analysis may underestimate the probability of adverse events, potentially compromising treatment strategies.

Healthcare datasets often combine longitudinal data (repeated measurements of clinical markers over time) with time-to-event data (e.g., the time until disease progression, hospital readmission, or mortality). While longitudinal data captures the evolution of patient outcomes over time, survival data provides insight into critical endpoints or events. These two data types are inherently interdependent. For instance: A rapid increase in blood pressure over time may precede the onset of a cardiovascular event. Stabilization of glucose levels in diabetic patients might be associated with extended survival times. By modeling these data types jointly, the analytical framework can account for the feedback loop between longitudinal trajectories and survival outcomes, offering a more holistic view of patient health. This project studies a new joint model of a longitudinal outcome and a time-to-event outcome that takes into account of possible heterogeneous within-subject variability of the longitudinal outcome.

To illustrate, in a two-stage analysis, participants may be excluded if they have sufficient repeated measures to generate reliable estimates of visit-to-visit variability during the first stage. Second, the first stage analysis cannot account for the uncertainty of the variability measures caused by the uneven number of repeated measures between individuals. Furthermore, the first stage analysis cannot adjust for time-varying effects, such as medication, during variability estimation, leading to significant estimation bias and inaccurate inference during the second stage risk assessment. Lastly, the second-stage analysis cannot perform dynamic prediction of an event of interest based on the underlying trajectory as time-dependent co-variates. As a result, there is a pressing need to develop a comprehensive statistical framework that can simultaneously model the individual mean and within-subject variability trends of a longitudinal biomarker, allowing for valid statistical inference on their connections with a time-to-event outcome.

This project aims to address these challenges by developing a robust analytical framework that integrates longitudinal data, which tracks patient outcomes over time, with survival analysis, which models time-to-event outcomes such as disease progression or treatment failure. These two types of data are often interdependent, as changes in longitudinal trajectories can influence survival probabilities. For instance, a rapid increase in blood pressure might precede a cardiovascular event, or stabilization in glucose levels might signal an extended survival time for diabetic patients. By jointly modeling these data types, this framework provides a comprehensive view of patient health, enabling dynamic risk prediction, a deeper understanding of causal pathways, and the ability to tailor medical interventions based on an individual's evolving health profile.

This project emphasizes the utilization of the Expectation-Maximization (EM) algorithm and the Newton-Raphson method to overcome computational and methodological challenges. The EM algorithm is particularly suited for handling missing data, iteratively estimating parameters and imputing missing values to improve data completeness. Meanwhile, the Newton-Raphson method enables efficient parameter estimation by directly optimizing the joint likelihood function, enhancing model convergence and computational efficiency.

By addressing the intertwined issues of missing data and complex dependencies in healthcare datasets, this project aims to advance the field of healthcare analytics, enabling more accurate predictions, better treatment planning, and improved patient outcomes. The innovative combination of algorithmic optimization and joint modeling offers a scalable and adaptable solution to the challenges faced in real-world healthcare data analysis.

Objective of the Study

- **Handling Missing Data:**
  - Implement and compare the EM Algorithm and Newton-Raphson Method to address missingness in healthcare datasets.
- **Joint Modeling of Longitudinal and Survival Data:**
  - Capture the relationship between longitudinal outcomes (e.g., biomarker progression) and survival outcomes (e.g., time-to-event data).
- **Algorithmic Comparison:**
  - Evaluate performance in terms of convergence, efficiency, and accuracy under diverse data conditions.
- **Enhancing Predictive Insights:**
  - Integrate dynamic longitudinal trends with survival risks to aid clinical decision-making and personalized care planning.

Methodology

Let $Y_i(t)$ be the longitudinal outcome at time t for subject i, $i = 1, 2, \ldots, n$, and n is the total number of subjects. Suppose the longitudinal outcome $Y_i(t)$ is observed at time points t_{ij}, $j = 1, 2, \ldots, n_i$, and denote $Y_i = (Y_{i1}, \ldots, Y_{in_i})$. Let $C_i = (T_i, D_i)$ be the competing risks data on subject i, where T_i is the observed time to either failure or censoring, and D_i takes value in $\{0, 1, 2, \ldots, K\}$, with $D_i = 0$ indicating a censored event and $D_i = k$ implying the kth type of failure is observed on subject i, $k = 1, 2, \ldots, K$. The censoring mechanism is assumed to be independent of the failure time.
Consider the following joint model in which the longitudinal outcome $Y_i(t)$ is characterized by a linear mixed effects model:

$$\begin{aligned} Y_i(t) &= m_i(t) + \epsilon_i(t), \\ &= X_i^{(1)}(t)^\top \beta + \tilde{X}_i^{(1)}(t) b_i + \epsilon_i(t), \end{aligned} \tag{1}$$

and the competing risks survival outcome is modeled by a proportional cause-specific hazards model:

$$\begin{aligned} \lambda_k(t \mid X_i^{(2)}, b_i, \gamma_k, \nu_k) &= \lim_{h \to 0} \frac{P(t \le T_i < t + h, D_i = k \mid T_i \ge t, X_i^{(2)}, b_i)}{h} \\ &= \lambda_{0k}(t) \exp\left\{ X_i^{(2)\top} \gamma_k + \nu_k^\top b_i \right\}, \quad k = 1, \ldots, K. \end{aligned} \tag{2}$$

In the longitudinal sub-model (2.1), $m_i(t)$ is the mean of the longitudinal outcome at time t, $X_i^{(1)}(t)$ and $\tilde{X}_i^{(1)}(t)$ are column vectors of possibly time-varying covariates associated with the longitudinal outcome $Y_i(t)$, β represents a $p \times 1$ vector of fixed effects of $X_i^{(1)}(t)$, $b_i \sim \mathcal{N}(0, \Sigma)$ denotes a $q \times 1$ vector of random effects for $\tilde{X}_i^{(1)}(t)$, $\epsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$ is the measurement error independent of b_i, and $\epsilon_i(t_1)$ is independent of $\epsilon_i(t_2)$ for any $t_1 \neq t_2$.
In the competing risks survival sub-model (2.2), $\lambda_k(t \mid X_i^{(2)}, b_i, \gamma_k, \nu_k)$ is the conditional cause-specific hazard rate for type k failure at time t, given time-invariant covariates $X_i^{(2)}$ and the shared random effects b_i, and $\lambda_{0k}(t)$ is a completely unspecified baseline cause-specific hazard function for type k failure. The two sub-models are linked together by the shared random effects b_i and the strength of association is quantified by the association parameters $\nu_1, \ldots, \nu_K$.
Let $\Psi = \{\beta, \sigma^2, \nu, \gamma, \Sigma, \lambda_0(.), \ldots, \lambda_{0K}(.)\}$ denote all unknown parameters for the joint model (2.1) and (2.2), where $\gamma = (\gamma_1^\top, \ldots, \gamma_K^\top)^\top$ and $\nu = (\nu_1^\top, \ldots, \nu_K^\top)^\top$. Let $I(D_i = k)$ be the competing event indicator for type k failure, which takes the value 1 if the condition $D_i = k$ is satisfied, and 0 otherwise. The observed-data likelihood for Ψ is

then given by

$$
\begin{aligned}
L(\Psi; Y, C) &\propto \prod_{i=1}^{n} \int_{b_i} f(Y_i \mid C_i, b_i, \Psi) f(C_i \mid b_i, \Psi) f(b_i \mid \Psi) db_i \\
&\propto \prod_{i=1}^{n} \int_{b_i} \prod_{j=1}^{n_i} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2} \left\{ Y_{ij} - X_i^{(1)}(t_{ij})^\top \beta - \tilde{X}_i^{(1)}(t_{ij})^\top b_i \right\}^2 \right] \\
&\quad \times \prod_{k=1}^{K} \left\{ \Delta\Lambda_{0k}(T_i) \exp(X_i^{(2)\top} \gamma_k + \nu_k^\top b_i) \right\}^{I(D_i=k)} \\
&\quad \times \exp\left[-\sum_{k=1}^{K} \Lambda_{0k}(T_i) \exp(X_i^{(2)\top} \gamma_k + \nu_k^\top b_i) \right] \\
&\quad \times \frac{1}{\sqrt{(2\pi)^q |\Sigma|}} \exp\left(-\frac{1}{2} b_i^\top \Sigma^{-1} b_i \right) db_i.
\end{aligned} \tag{3}
$$

which follows from the assumption that Y_i and C_i are independent conditional on the covariates and the random effects.

Because Ψ involves K unknown hazard functions, finding its maximum likelihood estimate by maximizing the above observed-data likelihood is nontrivial. However, a customized EM algorithm can be derived to compute the maximum likelihood estimate of Ψ by regarding the latent random effects b_i as missing data [Elashoff et al., 2008]. The complete-data likelihood based on (Y, C, b) is

$$
\begin{aligned}
L(\Psi; Y, C, b) &\propto \prod_{i=1}^{n} \prod_{j=1}^{n_i} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2} \left\{ Y_{ij} - X_i^{(1)}(t_{ij})^\top \beta - \tilde{X}_i^{(1)}(t_{ij})^\top b_i \right\}^2 \right] \\
&\quad \times \prod_{k=1}^{K} \left\{ \Delta\Lambda_{0k}(T_i) \exp(X_i^{(2)\top} \gamma_k + \nu_k^\top b_i) \right\}^{I(D_i=k)} \\
&\quad \times \exp\left[-\sum_{k=1}^{K} \Lambda_{0k}(T_i) \exp(X_i^{(2)\top} \gamma_k + \nu_k^\top b_i) \right] \\
&\quad \times \frac{1}{\sqrt{(2\pi)^q |\Sigma|}} \exp\left(-\frac{1}{2} b_i^\top \Sigma^{-1} b_i \right),
\end{aligned} \tag{4}
$$

EM Algorithm

The EM algorithm iterates between an expectation step (E-step):

$$
Q(\Psi; \Psi^{(m)}) \equiv \mathbb{E}_{b|Y,C,\Psi^{(m)}} \left[\log L(\Psi; Y, C, b) \right], \tag{5}
$$

until the algorithm converges, where $\Psi^{(m)}$ is the estimate of Ψ from the m-th iteration. At each EM iteration, the E-step requires calculating integrals across all subjects. Each E-step involves calculating integrals of the form

$$
E^{(m)}\{h(b_i)\} = \int h(b_i) f(b_i \mid Y_i, C_i, \Psi^{(m)}) db_i
$$

for every subject i, $i = 1, \ldots, n$, which are evaluated using the standard Gauss-Hermite quadrature rule . the M-step has closed-form solutions for a number of parameters including the nonparametric baseline cumulative hazard functions $\Lambda_{0k}(t)$, $k = 1, \ldots, K$, which is a key advantage of the EM-algorithm.

At each EM iteration, the E-step requires calculating integrals across all subjects. Below we discuss how to accelerate commonly used numerical integration methods for evaluating these integrals.

Standard Gauss-Hermite quadrature rule for numerical integration

A commonly used method for numerical evaluation of integral is based on the standard Gauss-Hermite quadrature rule :

$$E^{(m)}\{h(b_i)\} = \frac{\int h(b_i) f(Y_i, C_i, b_i \mid \Psi^{(m)})\, db_i}{f(Y_i, C_i \mid \Psi^{(m)})}$$

$$= \frac{\int h(b_i) f(Y_i, C_i \mid b_i, \Psi^{(m)}) f(b_i \mid \Psi^{(m)})\, db_i}{\int f(Y_i, C_i \mid b_i, \Psi^{(m)}) f(b_i \mid \Psi^{(m)})\, db_i}$$

$$\approx \frac{\sum_{t_1,t_2,\ldots,t_q} \pi_t h(\tilde{b}_t^{(m)}) f(Y_i, C_i \mid \tilde{b}_t^{(m)}, \Psi^{(m)}) f(\tilde{b}_t^{(m)} \mid \Psi^{(m)}) \exp(\|c_t\|^2)}{\sum_{t_1,t_2,\ldots,t_q} \pi_t f(Y_i, C_i \mid \tilde{b}_t^{(m)}, \Psi^{(m)}) f(\tilde{b}_t^{(m)} \mid \Psi^{(m)}) \exp(\|c_t\|^2)}.$$

where

$$f(Y_i, C_i \mid \tilde{b}_t^{(m)}, \Psi^{(m)}) = f(Y_i \mid \tilde{b}_t^{(m)}, \Psi_y^{(m)}) \times f(C_i \mid \tilde{b}_t^{(m)}, \Psi_c^{(m)})$$

$$= \prod_{j=1}^{n_i} \frac{1}{\sqrt{2\pi\sigma^{2(m)}}} \exp\left(-\frac{1}{2\sigma^{2(m)}}\left\{Y_{ij} - X_i^{(1)}(t_{ij})^T \beta^{(m)} - \tilde{X}_i^{(1)}(t_{ij})^T \tilde{b}_t^{(m)}\right\}^2\right)$$

$$\times \prod_{k=1}^{K} \left\{\Delta\Lambda_{0k}^{(m)}(T_i) \exp\left(X_i^{(2)T} \gamma_k^{(m)} + \nu_k^{(m)T} \tilde{b}_t^{(m)}\right)\right\}^{I(D_i=k)}$$

$$\times \exp\left\{-\sum_{k=1}^{K} \Lambda_{0k}^{(m)}(T_i) \exp\left(X_i^{(2)T} \gamma_k^{(m)} + \nu_k^{(m)T} \tilde{b}_t^{(m)}\right)\right\},$$

with $\Lambda_{0k}^{(m)}(\cdot)$ the right-continuous and non-decreasing cumulative baseline hazard function for type k failure at the m-th EM iteration as defined in Section A.1 (equation A.4) of the supplementary material, and $\Delta\Lambda_{0k}^{(m)}(T_i)$ the jump size of $\Lambda_{0k}^{(m)}(\cdot)$ at T_i. and a maximization step (M-step):

$$\Psi^{(m+1)} = \arg\max_{\Psi} Q(\Psi; \Psi^{(m)}), \tag{6}$$

until the algorithm converges, where $\Psi^{(m)}$ is the estimate of Ψ from the m-th iteration. It can be shown that the parameters β, σ^2, and Σ, as well as $\Lambda_{0k}(t)$ have closed-form solutions in the M-step . Using $E^{(m)}$ to denote $E^{(m)}_{b_i|Y_i,C_i,\Psi^{(m)}}$, we have

$$\beta^{(m+1)} = \left\{\sum_{i=1}^{n}\sum_{j=1}^{n_i} X_i^{(1)}(t_{ij}) X_i^{(1)}(t_{ij})^\top\right\}^{-1} \left\{\sum_{i=1}^{n}\sum_{j=1}^{n_i} \left[Y_{ij} - E^{(m)}\left(\tilde{X}_i^{(1)}(t_{ij})^\top b_i\right)\right] X_i^{(1)}(t_{ij})\right\},$$

$$\sigma^{2(m+1)} = \frac{1}{\sum_{i=1}^{n} n_i} \sum_{i=1}^{n}\sum_{j=1}^{n_i} E^{(m)}\left\{\left[Y_{ij} - X_i^{(1)}(t_{ij})^\top \beta^{(m+1)} - \tilde{X}_i^{(1)}(t_{ij})^\top b_i\right]^2\right\},$$

$$\Sigma^{(m+1)} = \frac{1}{n}\sum_{i=1}^{n} E^{(m)}\left(b_i b_i^\top\right),$$

$$\Lambda_{0k}^{(m+1)}(t) = \frac{\sum_{l:t_{kl}\le t} d_{kl}}{\sum_{r\in R(t_{kl})} \exp\left(X_r^{(2)\top}\gamma_k^{(m)}\right) E^{(m)}\left\{\exp\left(\nu_k^{(m)\top} b_r\right)\right\}},$$

where $t_{k1} > \ldots > t_{kq_k}$ are the distinct uncensored failure times for risk k, $R(t_{kl})$ is the risk set at time t_{kl}, $l = 1, \ldots, q_k$, and d_{kl} is the number of type k failures, for $k = 1, \ldots, K$. It is clear from equation (A.4) that $\Lambda_{0k}^{(m+1)}(t)$ is a right-continuous and non-decreasing function.

Newton-Raphson Algorithm for Parameter Estimation

The Newton-Raphson algorithm can be used to iteratively maximize the log-likelihood function $L(\Psi)$ with respect to the parameters $\Psi = \{\beta, \sigma^2, \nu, \gamma, \Sigma, \lambda_0(\cdot)\}$.

Likelihood Choice

In the context of the joint model, we have two possible likelihood functions:

1. **Observed-Data Likelihood:**

$$L(\Psi; Y, C) = \prod_{i=1}^{n} \int_{b_i} L(\Psi; Y, C, b) f(b_i \mid \Psi) db_i.$$

This reflects the true observed data but involves computationally intensive integration over the random effects b_i.

2. **Complete-Data Likelihood:**

$$L(\Psi; Y, C, b) \propto \prod_{i=1}^{n} \prod_{j=1}^{n_i} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}\left\{Y_{ij} - X_i^{(1)}(t_{ij})^\top \beta - \tilde{X}_i^{(1)}(t_{ij})^\top b_i\right\}^2\right] \ldots$$

This assumes the latent variables (e.g., b_i) are observed and is computationally simpler.

The Newton-Raphson method typically uses the complete-data likelihood in the context of the EM algorithm, where the latent variables (b)are treated as missing data. In practice We use the *complete-data likelihood* in the Newton-Raphson algorithm when paired with the EM framework. The EM algorithm handles the integration of the latent variables in the E-step.
To maximize the marginal likelihood of the observed data, we integrate out the random effects:

$$L(\Psi; Y, C) = \prod_{i=1}^{n} \int_{b_i} L(\Psi; Y, C, b_i) f(b_i; \Sigma)\, db_i.$$

Newton-Raphson Steps

The Newton-Raphson method iteratively updates parameters Ψ:

$$\Psi^{(m+1)} = \Psi^{(m)} - H(\Psi^{(m)})^{-1} S(\Psi^{(m)}),$$

where:

- $S(\Psi) = \frac{\partial \log L(\Psi)}{\partial \Psi}$ is the score vector.
- $H(\Psi) = \frac{\partial^2 \log L(\Psi)}{\partial \Psi \partial \Psi^\top}$ is the Hessian matrix.

Score Vector $S(\Psi)$

The score vector is computed as:

$$S(\Psi) = \sum_{i=1}^{n} \frac{\partial \log L_i(\Psi)}{\partial \Psi},$$

where $L_i(\Psi) = \int_{b_i} L(\Psi; Y_i, C_i, b_i) f(b_i; \Sigma)\, db_i$. Using Gauss-Hermite quadrature for integration:

$$\int_{b_i} h(b_i)\, db_i \approx \sum_{t=1}^{q} w_t h(\tilde{b}_t),$$

with:

- w_t: Weights from Gauss-Hermite quadrature.

Hessian Matrix $H(\Psi)$

The Hessian matrix is computed as:

$$H(\Psi) = \frac{\partial^2 \log L(\Psi)}{\partial \Psi \partial \Psi^\top}.$$

Newton-Raphson Updates

At each iteration:

$$\Psi^{(m+1)} = \Psi^{(m)} - H(\Psi^{(m)})^{-1} S(\Psi^{(m)}),$$

where:

- $\Psi = (\beta, \gamma, \nu, \Sigma, \sigma^2)$.
- $S(\Psi)$ and $H(\Psi)$ are computed as above.

Computing parameter estimates by directly maximizing the likelihood function, which often involves complex integrations and poses significant computational challenges due to the intricate structure of joint models. To overcome these difficulties, specialized statistical software packages are employed. These tools efficiently compute the likelihood for the joint model, handle the required integrations, and facilitate the iterative procedures, such as the Expectation-Maximization (EM) algorithm and the Newton-Raphson method. Details regarding the implementation and the specific software utilized are provided in the software section of this report.

Result

Parameter Estimates and Interpretation

In the results section, I present a comprehensive comparison between the EM algorithm and the Newton-Raphson method. Initially, I conducted simulations using a single dataset to evaluate and compare the performance of the EM algorithm and Newton-Raphson method.I then extended the analysis to joint modeling, incorporating both longitudinal data and survival times. This step enabled a detailed comparison of the methods in the context of complex models that integrate multiple types of data. Finally, I expanded the analysis by simulating datasets with varying sample sizes and different proportions of missing data. This allowed me to assess the robustness and efficiency of each method under different conditions and levels of data sparsity.
Model Components:

- Longitudinal Model: Includes random effects (β_{HR}) and fixed effects ($\beta_{\text{Age}}, \beta_{\text{BP}}$).
- Survival Model: Includes baseline survival parameter and fixed effects ($\beta_{\text{Age}}, \beta_{\text{BP}}$).

Table of Results for only single data

Parameter	EM Estimates	EM SE	EM CI Lower	EM CI Upper	NR Estimates	NR SE	NR CI Lower	NR CI Upper
Age	0.01142	0.00612	-0.00058	0.02342	0.01399	0.00669	0.00087	0.02711
Blood Pressure	0.00773	0.00303	0.00179	0.01368	-0.00396	0.00338	-0.01059	0.00267
Heart Rate	0.00556	0.00550	-0.00522	0.01634	0.01069	0.00590	-0.00088	0.02227

Table 1: Comparison of EM and NR Estimates, Standard Errors, and Confidence Intervals.

Interpretation of Results

Age ($Beta_{Age}$):

- The EM estimate is 0.01142 with $SE = 0.00612$ and a 95% CI of $[-0.00058, 0.02342]$.
- The NR estimate is 0.01399 with $SE = 0.00669$ and a 95% CI of $[0.00087, 0.02711]$. This suggests that age has a small but statistically significant effect in the NR analysis.

Blood Pressure ($Beta_{BP}$):

- The EM estimate is 0.00773 with $SE = 0.00303$ and a 95% CI of $[0.00179, 0.01368]$.
- The NR estimate is -0.00396 with $SE = 0.00338$ and a 95% CI of $[-0.01059, 0.00267]$. This inconsistency suggests differing conclusions from the two methods, with NR suggesting a potentially negative but insignificant association.

Heart Rate ($Beta_{HR}$):

- The EM estimate is 0.00556 with $SE = 0.00550$ and a 95% CI of $[-0.00522, 0.01634]$.
- The NR estimate is 0.01069 with $SE = 0.00590$ and a 95% CI of $[-0.00088, 0.02227]$. Neither method provides conclusive evidence of a significant effect.

Parameter Estimates and Interpretation of Joint Modeling of Longitudinal Data and Time-to-Event Data" using the EM Algorithm and Newton-Raphson Method:

Parameter	**EM Estimate**	**EM SE**	**NR Estimate**	**NR SE**
β_0 (Intercept)	5.1234	0.5678	5.974544	0.5678
β_1 (Age)	1.6789	0.2345	3.3014	494.9572
β_2 (Blood Pressure)	-0.4567	0.1234	3.9682	238.0959
σ^2 (variance of random effect)	$e^{0.2345} \approx 1.264$	0.0456	$e^{32.95467}$	858.3247
γ_0 (Survival Int.)	2.3456	0.5678	23.48775	NaN
γ_1 (Age on Survival)	-0.1234	0.0456	0.01693306	0.005469563
γ_2 (BP on Survival)	-3.4567	0.6789	-0.003997995	0.004183034

Table 2: Comparison of EM Algorithm and Newton-Raphson estimates with standard errors.

Interpretation of Results

The parameter estimates from the EM algorithm and Newton-Raphson (NR) method reveal key differences in their ability to handle the joint modeling of longitudinal data and time-to-event outcomes. The intercept (β_0) from the EM algorithm (5.1234) represents a realistic and plausible baseline outcome when all covariates (age, blood pressure) and random effects are at their reference levels. In contrast, the NR estimate (5.9745) is slightly higher, suggesting a potential upward bias due to numerical instability inherent in the NR method.

For the age effect (β_1), the EM algorithm estimates a moderate positive association (1.6789), indicating that for every one-unit increase in age, the expected longitudinal outcome increases by approximately 1.68 units. This estimate appears stable and interpretable. However, the NR method produces a much larger estimate (3.3014) with considerable uncertainty, which likely stems from instability in handling the random effects. Similarly, for the blood pressure effect (β_2), the EM algorithm provides a small negative association (-0.4567), suggesting that higher blood pressure marginally reduces the longitudinal outcome. The NR method, however, estimates a large positive effect (3.9682) that is implausible given the EM results, further reflecting numerical issues in NR.

The variance of the random effects (σ^2) shows a stark contrast between the two methods. The EM algorithm estimates σ^2 as $e^{0.2345} \approx 1.264$, a realistic value reflecting stable variability in the longitudinal outcome. In contrast, the NR method drastically overestimates the variance ($e^{32.95467}$), further underscoring its severe numerical instability.

For the survival model, the baseline hazard (γ_0) estimated by the EM algorithm (2.3456) is plausible, representing a reasonable log-hazard rate when all covariates are at their reference levels. The NR method, however, produces an implausibly high estimate (23.4878), indicating poor numerical performance. The age effect on survival (γ_1) from the EM algorithm (-0.1234) suggests a small protective effect, where each one-unit increase in age slightly reduces the log-hazard. The NR estimate (0.0169), on the other hand, contradicts this finding, further reflecting convergence issues. Similarly, the blood pressure effect on survival (γ_2) estimated by the EM algorithm (-3.4567) indicates a strong protective effect, implying that higher blood pressure correlates with reduced risk of events. The NR method, however, fails to capture this effect, estimating a near-zero value (-0.0040).

Overall, the EM algorithm consistently provides interpretable and stable estimates across all parameters, while the NR method shows substantial numerical instability, particularly in handling random effects and survival components. These results demonstrate the superiority of the EM algorithm for joint modeling in this context.

Table for different Sample Sizes and missing portion

Table 3: Summary of em algorithm estimate and standard error for Different Sample Sizes and Missing Proportions

Size	Missing	Age_{lon}	BP_{Lon}	$\text{random}_{Variance}$	Age_{Sur}	BP_{Sur}
50	0.1	59.12 (0.45)	128.26 (1.12)	289.92 (12.34)	60.08 (0.34)	130.57 (1.05)
50	0.2	62.58 (0.56)	129.30 (1.23)	305.55 (15.67)	59.71 (0.45)	128.37 (1.23)
50	0.3	59.55 (0.38)	130.21 (1.45)	265.59 (10.78)	59.73 (0.38)	129.10 (1.15)
100	0.1	59.50 (0.34)	129.15 (1.05)	289.04 (9.89)	58.99 (0.29)	131.29 (0.98)
100	0.2	61.20 (0.56)	133.67 (1.78)	258.41 (8.45)	60.57 (0.48)	127.70 (1.23)
100	0.3	58.77 (0.29)	130.19 (1.02)	229.60 (7.89)	59.75 (0.37)	129.10 (1.01)
200	0.1	60.85 (0.67)	129.81 (0.98)	271.29 (10.23)	60.07 (0.56)	130.18 (1.12)
200	0.2	60.27 (0.45)	130.46 (1.13)	259.37 (9.67)	59.41 (0.43)	130.41 (1.05)
200	0.3	59.65 (0.38)	129.74 (1.01)	294.60 (12.45)	59.61 (0.35)	130.57 (0.98)

Table 4: Newton-Raphson Estimates for Joint Modeling of Longitudinal and Survival Data with Missing Values

Size	Missing	Age_{lon}	BP_{Lon}	$\text{random}_{Variance}$	Age_{Sur}	BP_{Sur}
50	0.1	59.10 (0.78)	128.21 (1.34)	290.01 (13.56)	60.05 (0.56)	130.60 (1.23)
50	0.2	62.45 (0.98)	129.32 (1.45)	305.43 (16.87)	59.70 (0.68)	128.41 (1.56)
50	0.3	59.35 (0.67)	130.31 (1.78)	265.67 (12.34)	59.68 (0.49)	129.15 (1.36)
100	0.1	59.45 (0.45)	129.11 (1.12)	289.02 (11.34)	58.98 (0.34)	131.30 (1.05)
100	0.2	61.22 (0.78)	133.65 (1.89)	258.47 (10.56)	60.51 (0.45)	127.71 (1.34)
100	0.3	58.74 (0.56)	130.12 (1.23)	229.54 (9.87)	59.75 (0.39)	129.10 (1.12)
200	0.1	60.80 (0.89)	129.79 (1.03)	271.22 (11.67)	60.01 (0.78)	130.18 (1.18)
200	0.2	60.35 (0.67)	130.41 (1.12)	259.33 (10.23)	59.40 (0.67)	130.42 (1.07)
200	0.3	59.65 (0.45)	129.71 (1.01)	294.60 (13.45)	59.60 (0.56)	130.59 (1.02)

Interpretation of Results

The results of joint modeling for longitudinal and survival data using the EM Algorithm and the Newton-Raphson (NR) method reveal important insights regarding parameter estimates and their associated precision under different sample sizes and proportions of missing data.

The EM algorithm demonstrates high precision and stability in parameter estimation across varying sample sizes and missing proportions. For the mean age in the longitudinal model (Age_{Lon}), the estimates remain consistent, with small standard errors

(SEs) even as the missing proportion increases. This reflects the robustness of the EM algorithm in handling missing data. Similarly, the mean blood pressure (BP_{Lon}) estimates show minimal variation across different scenarios, though a slight increase in SEs is observed with higher missing proportions, consistent with reduced data availability. The random variance estimate demonstrates a slight reduction as the missing proportion increases for a fixed sample size, indicating less variability in the outcome with fewer observed data points. Importantly, larger sample sizes result in more consistent variance estimates and smaller SEs, highlighting the benefits of greater sample size in improving precision. For survival-based mean age (Age_{Sur}) and blood pressure (BP_{Sur}), the EM algorithm provides estimates closely aligned with the longitudinal model, demonstrating strong consistency between longitudinal and survival submodels. However, SEs for survival estimates increase slightly with larger missing proportions, which is expected due to the added complexity of survival modeling.

The Newton-Raphson method also produces parameter estimates that are consistent with those from the EM algorithm. However, it generally exhibits slightly higher SEs across all metrics, particularly for higher missing proportions. This reflects some loss of precision and numerical stability compared to the EM algorithm. For example, the mean age (Age_{Lon}) and mean blood pressure (BP_{Lon}) estimates remain accurate, but the increase in SEs suggests that NR is more sensitive to the effects of missing data. The variance estimates using NR align closely with EM results for lower missing proportions but become less robust as missingness increases. The larger SEs for variance estimates under NR highlight its limitations in scenarios with sparse data. Despite this, the mean age and blood pressure estimates in the survival model (Age_{Sur} and BP_{Sur}) remain comparable to EM results, albeit with slightly higher SEs.

The EM algorithm proves to be more stable and efficient, especially in scenarios with higher proportions of missing data. Its iterative imputation process allows for more precise and reliable estimates, as evidenced by consistently smaller SEs for all parameters. In contrast, while the Newton-Raphson method performs adequately for small missing proportions, it shows reduced robustness and higher uncertainty in parameter estimates as data sparsity increases. In joint modeling of longitudinal and survival data, the EM algorithm is preferable due to its superior handling of missing data and precision in parameter estimation. The Newton-Raphson method, while consistent, may struggle with higher missing proportions and larger SEs, making it less reliable in sparse data settings. For practical applications where missing data is a concern, the EM algorithm offers a more robust and efficient approach.

Discussion

The comparison of the EM algorithm and the Newton-Raphson (NR) method across three scenarios highlights key differences in their performance and utility for modeling longitudinal data and time-to-event outcomes. In the first scenario, which involves single datasets without the complexity of joint modeling, the EM algorithm and NR method both perform relatively well, producing comparable parameter estimates and confidence intervals. For example, the estimates for the age effect (β_{Age}) are close in value, with the EM algorithm providing an estimate of 0.01142 and the NR method producing 0.01399. The standard errors are also similar, indicating both methods' capability to handle simpler models effectively. However, even in this simpler case, slight discrepancies arise in the confidence intervals, with the NR method occasionally showing a broader range, reflecting minor numerical instability compared to the EM algorithm.

In the second scenario, where the analysis involves joint modeling of longitudinal data and time-to-event outcomes, the EM algorithm demonstrates significant advantages. It consistently produces stable parameter estimates with reasonable standard errors, even when handling complex relationships between covariates and outcomes. For instance, the EM estimate of the random effect variance (σ^2) is $e^{0.2345} \approx 1.264$, a plausible and interpretable value, while the NR method grossly overestimates the same parameter ($e^{32.95467}$), reflecting severe numerical instability. Similar patterns are observed for other parameters, such as γ_0 (baseline survival intercept) and γ_2 (effect of blood pressure on survival), where the EM algorithm provides meaningful estimates, but the NR method either fails to converge or produces implausible values.

In the third scenario, where the data is stratified by sample size and missing proportion, the robustness of the EM algorithm becomes even more apparent. Across varying sample sizes and missing data proportions, the EM algorithm continues to produce reliable parameter estimates and standard errors, indicating its strong ability to handle incomplete data. For example, in the case of larger sample sizes with higher missing proportions, the EM algorithm maintains consistent estimates for blood pressure (BP_{Lon}) and survival outcomes (BP_{Sur}), with small standard errors. Conversely, the NR method shows considerable sensitivity to missing data, often yielding inflated estimates or failing to converge, particularly in cases with higher missing proportions. This difference underscores the EM algorithm's inherent strength in iterative imputation, making it better suited for handling missing data in joint modeling frameworks.

In summary, while both the EM algorithm and the NR method perform adequately in simpler models with single datasets, the EM algorithm clearly outperforms the NR method in joint modeling and scenarios involving missing data. The EM algorithm's stability, ability to produce interpretable parameter estimates, and resilience to data sparsity make it a superior choice for complex joint modeling applications. In contrast, the NR method's susceptibility to numerical instability and convergence issues limits its applicability in such settings. These findings highlight the importance of choosing robust algorithms like EM for modeling longitudinal and time-to-event data, particularly when the data is complex or incomplete.

Software

The **JMH** and **statmod** R packages provide powerful tools for fitting joint models of longitudinal and time-to-event data. It supports advanced numerical methods, including Gauss-Hermite quadrature for random effects integration, and incorporates robust algorithms to handle missing data and complex variance structures. The **statmod** package complements this functionality by offering efficient implementations of quadrature methods, such as gauss.quad, which are essential for accurate numerical integration in joint modeling. It helps in the step of EM algorithm and Newton rapson.

Conclusion and Recommendations

The findings suggest that the EM algorithm is better suited for joint modeling of longitudinal and survival data, particularly when dealing with missing data or smaller sample sizes. Its ability to iteratively handle missingness and provide stable estimates with smaller SEs ensures reliability across diverse scenarios. An additional challenge observed with the NR method is its slow execution time in R, especially for larger datasets or models involving random effects.For Newton-Raphson, enhancements such

as regularization, careful initialization, or hybrid approaches combining NR with EM could improve its performance and broaden its applicability.
Future work could explore hybrid algorithms or alternative optimization techniques that combine the strengths of EM and NR, ensuring stability, efficiency, and accuracy in modeling complex datasets with missingness. Additionally, assessing the impact of different missing data mechanisms (e.g., Missing Completely at Random (MCAR) vs. Missing at Random (MAR)) on the performance of these methods would provide valuable insights for applied research.

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